

Sarah Agrawal

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EDUCATION

University of California, Berkeley

Berkeley, CA

B.A. in Computer Science & Applied Mathematics

Expected May 2028

- Relevant Coursework: Data Structures & Algorithms, Artificial Intelligence, Full Stack Web Development, Structure & Interpretation of Computer Programs, Internet Architecture & Protocols, Principles and Techniques of Data Science, Discrete Mathematics, Linear Algebra & Differential Equations

TECHNICAL SKILLS

Languages: Java, Python, JavaScript/TypeScript, SQL, HTML/CSS, Scheme

Frameworks & Libraries: Jetty, StdDraw, scikit-learn, Pandas/NumPy

Developer Tools: Git/GitHub, REST APIs, JUnit, GPT-4, Claude Code

EXPERIENCE

Data Science & AI Intern

May 2026 – Present

Filmcrowd

- Architecting a **predictive-modeling workflow** (feature engineering, model training, importance ranking) to evaluate how budget, genre, and release-timing affect box-office performance
- Compiled and cleaned a dataset of production and release records for **1M+ films**, engineering reusable **feature-extraction modules** and running **hypothesis tests** to identify the factors driving box-office performance
- Conducted and recorded interviews with industry professionals – including producer **Barrie Osborne** (*The Lord of the Rings*, *The Matrix*, *The Great Gatsby*) and screenwriter **Charles Erven** (*Mulan*, *Big Man*) – compiling a proprietary **transcript dataset** used to inform model development

Software/AI Intern

Feb 2024 – Apr 2024

Bare Cove Technology

- Authored a technical research paper on **generative-AI** applications in **cybersecurity**, presenting findings and recommendations directly to the CEO
- Prototyped **GenAI-driven automation** for security operations workflows, pinpointing opportunities to cut manual analyst workload

Machine Learning Researcher

Dec 2023 – Apr 2024

Carnegie Mellon University

- Engineered an end-to-end **ML pipeline** in **Python** (data cleaning, feature engineering, model training) to classify **36,000+ labeled badminton shots** by outcome using a **Random Forest classifier**
- Built an automated **cross-validation** and **feature-importance evaluation** framework to assess model robustness, translating diagnostics into data-driven strategy recommendations for coaching staff

PROJECTS

BYOW | Java, StdDraw, procedural generation

Jul 2026

- Built a **procedural world-generation algorithm** that carves connected rooms and turning hallways from a seeded RNG, guaranteeing full floor-tile reachability and exact seed-to-seed reproducibility
- Implemented a **real-time tile-rendering engine** and game loop with keyboard-driven interactivity, a live HUD, and full **save/load state serialization** for exact session restoration
- Extended the engine with ambition features, including dynamic lighting and a **procedural music system**, to enhance immersion within the performance-sensitive rendering loop

Ngordnet | Java, Jetty, custom graph & TreeMap-based data structures

Jun 2026

- Designed a **TreeMap-backed TimeSeries** type and a custom **NGramMap** indexing engine that parses a **300MB Google Books NGrams corpus** in a single pass, enabling fast repeated word-frequency lookups without re-reading data on each query
- Built a **directed graph class** and **WordNet parser** from scratch, using graph traversal to resolve hyponym/hypernym relationships and support multi-word set-intersection queries with frequency-ranked top-k filtering
- Shipped a **Jetty-based REST API** layer connecting the Java backend to a JS/HTML frontend, enabling interactive word-history and synset visualizations end-to-end

TurfFlux | Python, Jupyter, EcoSIM

Mar – May 2026

- Created a **Python pipeline** that parses and translates **Dynamic Energy Budget (DEB)**-derived microbial trait parameters into EcoSIM's input format, automating a step that previously required manual reformatting
- Configured and scripted automated runs of **EcoSIM**, a process-based terrestrial ecosystem model, to simulate **N₂O emissions** from UC Berkeley's managed lawns through an independent research fellowship with BRLOC (Berkeley Research and Leadership Opportunities for Change)